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Hyper Text Markup Language

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What is an HTML File?

- An HTML file is a text file containing small markup tags.
- The markup tags tell the web browser how to display the page.
- An HTML file must have an **htm** or **html** file extension.
- An HTML file can be created using a simple text editor.

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Text Editors

- Creating a website requires:
  - A text editor – NotePad++, Atom, Sublime Text, Vi, *etc.*
  - There are other tools known as web authoring tools.
- A web hosting service.
- A browser

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HTML Tag Template

```
<html>
  <head>
    <title>
      (insert text for the browser's title bar here)
    </title>
  </head>
  <body>
      (insert visible Web page elements here)
  </body>
</html>
```

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An HTML Example

```
<html>
   <head>
        <title> Welcome to Web Programming </title>
   </head>
   <! - - this is a simple example - - >
   <body>
        <h2> This is the first lecture </h2>
        <p> We are going to revise basic HTML and introduce
   advanced HTML. </p>
        <hr> </hr>
   </body>
</html>
```

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Basic HTML

- HTML Basic Tags
- Formatting
- Entities
- Links
- Frames
- Tables
- Lists
- Forms
- Images
- Background
- Checkbox
- Radio Button
- Button

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HTML Basics

- HTML is composed of tags. HTML tags are always enclosed in angle brackets, **<>**, and are case-insensitive.
- Tags typically occur in begin-end pairs, with an opening and closing tag.
  - <tag></tag>
- Web pages begins with **<html>** and ends with **</html>**.

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Head and Title Tags

- The HEAD tags contain all of the document's header information.
  - <head></head>
  - It doesn't appear at the top of the browser window, but contains things like the document title and so on.

- The Title tag is placed within the HEAD structure. Between the TITLE tags, you should have the title of your document. This will appear at the top of the browser's title bar.
  - <title></title>

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Body Tag

- The BODY tags come after the HEAD tags. Between the BODY tags you find all of the stuff that gets displayed in the browser window. All of the text, the graphics, and links, and so on.
  - <body></body>

- Placing a background image can be done by typing in the following:
  - <body background="myImage.jpg" text="white">

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Webpage

- So, putting everything we've covered this far into one file we have:

```
<html>
    <head>
    <title>My First Webpage </title>
    </head>

    <body bgcolor="cyan" text="red">
    </body>

</html>
```

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Headings and Paragraphs

- There are six levels of headings, from heading 1 **<h1>** through to heading 6 **<h6>**.
- The beginning of a paragraph is marked by ** **, and the end by ** **.

 This is a paragraph.

 This is a paragraph .

 This is a paragraph.

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Text Formatting

- If you want to end a line after a certain word, but don’t want to start a new paragraph, you can use the **
** tag.

- Text on a web page can be set to **bold**, *italic*, and <u>underline:</u>
  - - is for **bold**.
  - - is for *italics*.
  - <u></u> - is for *underline*.

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Font Tag

- The FONT and COLOR tags can be used to change the fonts and colours of your web page.
  - <font face = “Arial” color = “blue”> Your Text </font>
- The Horizontal Rule <hr /> is useful when you want to separate two sections with a line.

This is a paragraph

This is a paragraph

This is a paragraph

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Hyperlinks

- In order to create a hyperlink, you'll need to know two things.
  1. The URL – the location of the file to which you want to link to.
  2. How links works.
- Links work by using the HREF (**H**ypertext **REF**erence) tag and the URL you want to link to.
  - <a href=http://www.google.com/> Google</a>
- Note: **a** signifies the anchor.

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Lists

- There are two types of lists in HTML – Ordered lists, and unordered lists.
  - <ol> and </ol> for ordered lists (numbered).
  - <ul> and </ul> for unordered lists (bullets).
  - <li> and </li> for items on the list.

```
<ol>
  <li>Monday</li>
  <li>Tuesday </li>
  <li>Wednesday </li>
  <li>Thursday </li>
  <li>Friday</li>
</ol>
```

would produce

1. Monday
2. Tuesday
3. Wednesday
4. Thursday
5. Friday

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Images

- Images are placed in web documents using the IMG tag.
- In order to make the IMG tag work, you need to use an SRC attribute.
  - SRC stands for source, where SRC is the URL of the graphic you want to have displayed.
- <img src="image.jpg" width="400" height="300">

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Tables

- The Table tag consists of the <table> and </table> tags.
  - <caption></caption> - used to create the title of a table.
  - <tr></tr> - used to create a table row.
  - <th></th> - used to create a header cell.
  - <td></td> - used to create a table cell.
  - Border – specify the border size.
  - Bordercolor – specify the colour of the border.
  - Align – specify the alignment of the whole table.
  - Cellspacing – specify the amount of empty space between the outer edges of each table cell.
  - Cellpadding – specify the amount of empty space between the border of the table cell and the content of the cell.

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Table

```
<table border="3" bordercolor="blue"
cellspacing="10" cellpadding="10"
align="center">
<caption>simple table </caption>
<tr>
   <td>row 1/column 1</td>
   <td>row 1/column 2</td>
   <td>row 1/column 3</td>
</tr>
<tr>
   <td>row 2/column 1</td>
   <td>row 2/column 2</td>
   <td>row 2/column 3</td>
</tr>
</table>
```

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Tables

- Tables can be used to organise the layout of your webpage.

```
<table width="60%" border="1">
<caption>another simple table </caption>
<tr>
   <td width="30%">row 1/column 1</td>
   <td width="70%">row 1/column 2</td>
</tr>
<tr>
   <td>row 2/column 1</td>
   <td>row 2/column 2</td>
</tr>
</table>
```

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Tables

- Rows can be expanded.

```
<table width="500" border="1">
<caption>expanding table </caption>
<tr>
   <td width="100">row 1/column 1</td>
   <td width="400">row 1/column 2</td>
</tr>
<tr>
   <td rowspan="2">row 2/column 1</td>
   <td>row 2/column 2</td>
</tr>
<tr>
   <td>row 3/column 2</td>
</tr>
</table>
```

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Tables

- Columns can also be expanded.

```
<table width="500" border="1">
<caption>expanding table </caption>
<tr>
   <td colspan="2">row 1/column 1</td>
   <td>row 1/column 3</td>
</tr>
<tr>
   <td>row 2/column 1</td>
   <td>row 2/column 2</td>
   <td>row 2/column 3</td>
</tr>
</table>
```

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Forms

- A form can contain input elements like text fields, checkboxes, radio-buttons, submit buttons *etc.*
- Forms are placed in web documents using the **form** tag.
  - <form name="InputTypes">

- The input tag is used to place input elements on the web page.
  - <input type="text" name="FirstName" size="20">

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Questions?